PRAVEENRAJ V

Aeronautical Engineer

+91 6383561935 • praveenraj.222004@gmail.com • Erode, Tamil Nadu, India • LinkedIn Profile

PROFESSIONAL SUMMARY

A Graduate Engineer with practical experience in CFD using ANSYS Fluent, applied across UAV and aerodynamics-focused projects. Proficient in simulation setup, flow behaviour analysis, and interpreting aerodynamic performance trends. Recognised for technical contribution as part of a team, achieved AIR 22 in SAE Autonomous Drone Development Challenge 2025 and Best Performer in SAE Drone Development Challenge 2025. Keen to apply experience in CFD, aerodynamic and structural analysis, and engineering simulation within the aerospace and defence industry, with a focus on performance optimization and design validation.

EDUCATION

B.E. Aeronautical Engineering — Kumaraguru College of Technology, Coimbatore - CGPA: **7.99** *2022 – 2026*

PROFESSIONAL EXPERIENCE

R&D Intern — KCIRI – Kumaraguru College of Technology, Coimbatore Dec 2025 – May 2026

- Conducted UAV conceptual design and sizing using Raymer's Aircraft Design methodology, evaluating multiple configurations based on mission requirements and performance targets.
- Automated UAV geometry generation and VLM-based aerodynamic analysis using OpenVSP Python API, enabling rapid design iterations and parametric studies.
- Integrated aerodynamic sizing, propulsion analysis, and performance evaluation workflows to streamline the conceptual design process.
- Performed transient CFD simulations of an industrial pump in ANSYS Fluent, gaining hands-on experience in meshing, inflation layer generation, moving mesh techniques, and flow analysis.

PROJECTS

UAV Landing Gear — Structural Design & FEA *SAEINDIA Drone Development Challenge 2025*

- Designed a lightweight landing gear system for a 5 kg UAV using SolidWorks, incorporating part and assembly modelling with reference-based design.
- Performed basic FEA in ANSYS Structural analysis. Gained practical exposure to structural response under loading conditions and supported geometry iterations aimed at improving the strength-to-weight ratio.

Computational Analysis — Dragonfly-Inspired Ornithopter (CFD) *Bioinspired Aerodynamics Study*

- Improved aerodynamic efficiency by 2.6% through transient CFD analysis of a dragonfly-inspired ornithopter using ANSYS Fluent.
- Performed geometry preparation, mesh generation, local sizing, inflation layer setup, and boundary condition definition for aerodynamic simulations.
- Conducted dynamic mesh analysis to evaluate real-world flapping-wing aerodynamic performance and support design improvements.

SKILLS:

CAD & Design: SolidWorks | OpenVSP | SpaceClaim | Catia
CFD & Analysis: ANSYS Fluent | External flow Analysis | Structural Analysis (ANSYS)
UAV Systems: Multirotor UAVs | Fixed-Wing UAVs | Payload Integration
Technical Tools: Microsoft Excel | Microsoft Word | Technical Documentation

ACHIEVEMENTS & LEADERSHIP

Technical Program Coordinator – Aeromodeling Club, Kumaraguru College of Technology

- Led the planning and execution of aerospace and UAV-focused technical events, workshops, and innovation challenges for students and industry participants 2025.
- Organized a week-long Drone Workshop and a 5-day Agricultural Drone Workshop, promoting practical applications of UAV technology and skill development 2024 & 2025.
- Coordinated a 12-hour UAV Ideathon and managed flagship Yugam 2025 events, including Drone Hunt, AUDC, and Drone Prix, driving participant engagement, team collaboration, and event operations.
- Achieved AIR 22 in the SAEINDIA Autonomous Drone Development Challenge (ADDC) 2025 for technical excellence in UAV design and development.
- Secured AIR 27 Overall and was recognized as Best Performer in the SAEINDIA Drone Development Challenge (DDC) 2025.
- Awarded for Leadership & Volunteerism, recognizing outstanding contribution to team management, coordination, and extracurricular initiatives 2025.
- Recognized for excellence in academic performance and co-curricular achievements 2025.